# MISS MINUTES & JARVIS AI ASSISTANT

Production Setup, Multilingual Architecture & Multi-LLM User Manual

## 1. Executive Overview & Architecture

Miss Minutes & Jarvis is a state-of-the-art dual-agent desktop assistant designed for Windows 10/11. The system coordinates two synchronized personas: Miss Minutes (warm, lively, user-facing TVA clock companion) and Jarvis (authoritative, technically precise capability architect). They feature sequential turn-taking dialogue, fluent bilingual comprehension in Telugu and English, hands-free voice activity detection, and deep desktop capability execution.

## 2. Turn-Taking Dialogue & Voice Intelligence

Miss Minutes and Jarvis communicate collaboratively without overlapping speech. Miss Minutes acknowledges the user's intent and provides warm conversational rapport, while Jarvis executes the underlying technical actions and provides verification.

- **Hands-Free Continuous Listening**: Continuous RMS-energy Voice Activity Detection (VAD) with 0.85s silence termination. No manual mic clicks required.
- **Quiet / Standby Mode**: Voice triggers like 'remain quiet', 'silent ga undu', or 'aagu' place the assistants in standby until asked to resume.
- **Neural Edge-TTS Voices**: Miss Minutes uses *te-IN-ShrutiNeural* (expressive Telugu voice) and Jarvis uses *te-IN-MohanNeural* (commanding male voice).
- **Zero-Lag Audio Caching**: Frequently spoken acknowledgments and greetings are cached in memory for 0ms playback latency.

## 3. Multi-Provider LLM Intelligence Backbone

The intelligence layer operates a resilient, self-healing chain across leading proprietary and open-source models:

| Provider | Models | Key Features |
|---|---|---|
| **Google Gemini** | gemini-2.5-flash, gemini-2.0-flash | Sub-second multimodal reasoning, native Telugu comprehension, structured JSON output. |
| **OpenAI & Compatible** | gpt-4o, gpt-4o-mini, Ollama (localhost) | Industry standard structured tool calling; seamless local inference via Ollama or LM Studio. |
| **Groq Cloud** | llama-3.3-70b-versatile, llama-3.1-8b | Ultra-high inference throughput (800+ tokens/sec) for zero conversational lag. |
| **Hugging Face Hub** | Llama-3.3-70B, Qwen-2.5-72B | Access to open-source foundation models via Serverless Inference API router. |
| **Free Public AI** | openai-fast, mistral | Key-free instant online LLM provider ensuring full functionality with zero setup. |

## 4. Contextual Adaptive Language & NLP Learning

The ContextLearner engine analyzes every interaction in real-time, adapting Miss Minutes and Jarvis to the user:

- **Vocabulary Extraction**: Tracks frequent Tenglish words ('workload', 'seconds lo', 'cheyyi', 'choodu') and mirrors them in conversation.

- **Personal Fact Memory**: Automatically records user details (name, tasks, preferences) and incorporates them across sessions.
- **Custom Shortcuts & Aliases**: Learns expressions such as 'wa ante whatsapp' or 'yt ante youtube' dynamically.
- **Persistent Profile**: Persists all learned context to ~/.miss_minutes/learned_context.json across reboots.

## 5. Desktop Capabilities & System Automation

- **Google Chrome**: Launch browser, open URLs, manage search queries and tabs.
- **WhatsApp**: Native desktop client protocol (whatsapp://) with instant fallback to web.whatsapp.com.
- **Screen Reader**: PySide6 primary display capture + Windows Media OCR to read questions or text off your screen.
- **Rapid Multi-Source Research**: Parallel queries to Wikipedia and DuckDuckGo with LLM synthesis in Telugu+English.
- **Security Boundary**: Default-DENY security model; destructive commands, shell scripts, and unauthorized paths are blocked.

## 6. Quick Start & Execution Commands

**1. Launch Desktop Assistant:**
python main.py or double-click launch_miss_minutes.bat

**2. Run Full System Diagnostics:**
python main.py --diagnostics

**3. Run Complete Test Suite (185 Tests):**
python -m pytest tests/